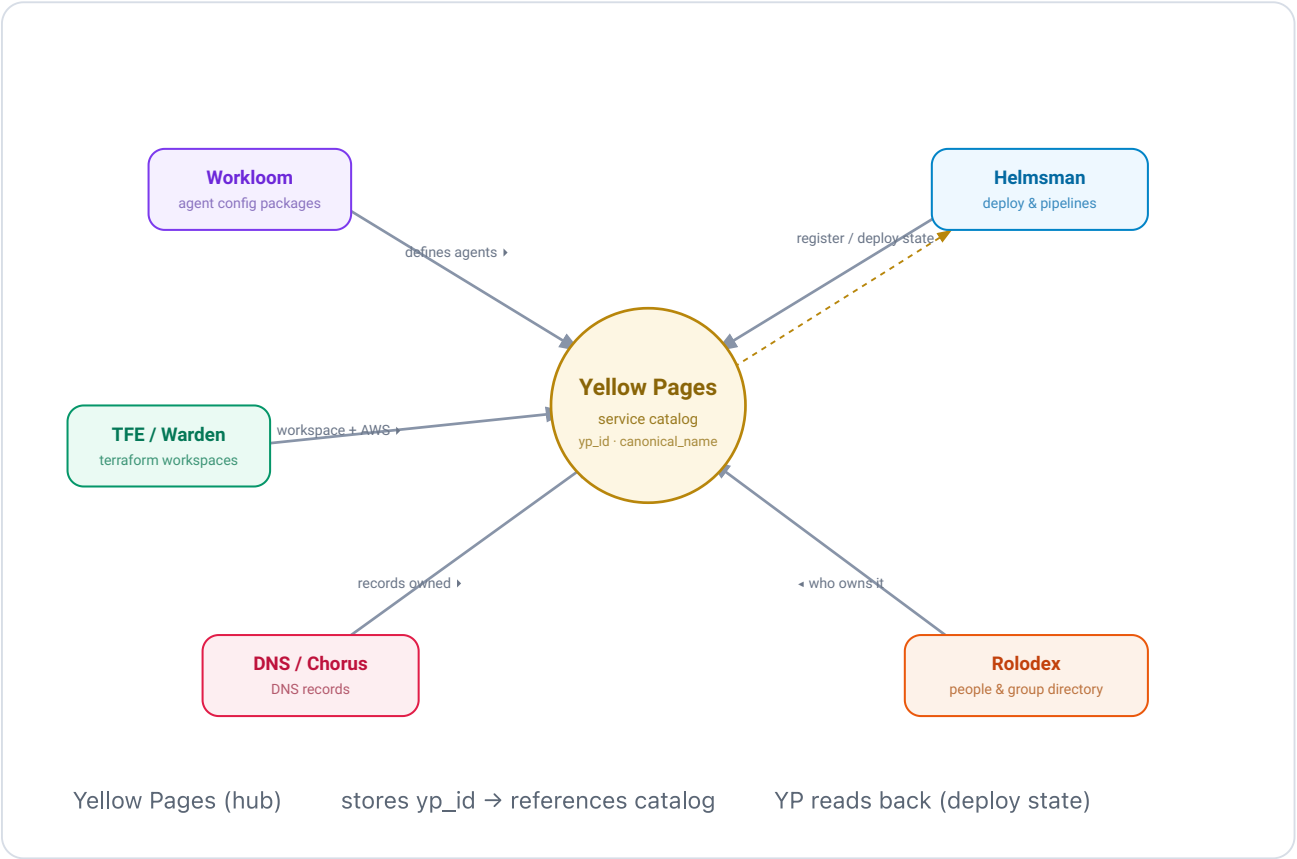


One Catalog to Bind Them

Six internal systems, one stack (Node 22 · Hono · Next.js · Postgres). **Yellow Pages** is the hub — every other system stores a Yellow Pages id and hangs off it. All hub and spoke; no sibling-to-sibling links.

The spine

Yellow Pages holds the source-of-truth ids. Everyone references them.



Who links how

Every edge is a loose reference — a stored id, not a hard SQL foreign key (the hub lives in a separate database).

Yellow Pages

shipped

THE HUB

Catalog of services, agents, teams, AWS/Azure accounts, labels. Exposes `_id` / `canonical_name`. Emits webhooks on change. Read-only connectors call siblings to surface live state.

Workloom

pull-side planned

AGENT SUPPLY CHAIN

Package manager for AI-agent config — one source rendered to Claude/Codex/Gemini/Copilot/Cursor. YP pins provenance via `definition_source` + `definition_ref` (e.g. `@org/refactor-agent@1.0.0`). YP adapters exist; the workloom search filter is not built yet.

Helmsman

shipped

BIDIRECTIONAL

Deploy/pipeline platform. `applications.yellowpages_service_id` links a deploy app to a catalog service. **Registers agents into YP** (`POST /api/agent/register`, idempotent). YP reads deploy state back = the observability view.

TFE / Warden

shipped

RESOURCE ALLOCATION

Terraform Cloud workspaces. Create takes `yp_service_id` + `yp_account_id`, pins a workspace to one service + one AWS account. Join tables map services to cloud accounts. That's how infra allocates to a catalog service.

DNS / Chorus

shipped

OWNERSHIP

Every DNS record carries a **required** `yellowpages_id` — records are owned by a catalog service. Unix `chmod`-style (9-bit) permissions layered on top. Trace any record back to its service.

Rolodex

shipped

IDENTITY RESOLUTION

Read-only LDAP/Workday/AD sync cache. Users & groups carry `yellowpagesServiceId`. Dedicated lookup `/user-by-yellowpages-service-id/{id}` resolves a service to real people/groups. Also SSH keys, PAT service accounts.

The agent lifecycle

Workloom → Yellow Pages → Helmsman. Half shipped, half aspirational — flagged honestly.

1

Define shipped

Workloom stores a versioned, trust-classed agent asset inside a package. Renders to every provider format.

2

Catalog shipped in YP

Yellow Pages records *where the agent lives*: `definition_source=workloom`, `definition_ref=@org/name@ver`. Provenance pinned.

3

Register shipped

Helmsman registers running agents back into YP (idempotent, loop-safe). YP now knows what's actually deployed.

4

Resolve & run planned (v2)

Helmsman fetches the pinned artifact bytes at deploy and runs it as an agentic workflow / builds an image. **Blocked**: no public byte-dereference endpoint in Workloom yet. Today it's a *pin*, not a running pipeline.

The "agent → docker image → agentic workflow" story is real intent but stops at step 3. Steps 1–3 are code; step 4 is design-locked, unbuilt.

Why hub-and-spoke wins

One id namespace turns six tools into one queryable system.



Single source of truth

Service identity lives in one place. No six copies drifting apart.



Observability for free

Ask YP one question, it fans out: deploy state, DNS, workspaces, owner — all keyed by `yp_id`.



Loose coupling

Refs are stored ids, not hard FKs. Any system can go down without breaking the others' schema.



Service → human

Rolodex maps a catalog service to the real people/groups who own it. Incident? You know who to page.



Safe sync

Idempotent registers + no-op webhook suppression + soft-archive-never-delete stop feedback loops.



Versioned provenance

Every agent traces to an exact `@scope/name@version`. Know precisely what's running.